

1. Introduction

This API documentation describes all backend endpoints provided by the **HandyChat** system. These APIs power:

- The HandyChat chatbot
- Student web UI
- Internal testing tools
- Future integrations (mobile app, WeChat mini-program, admin dashboard)

All API responses follow consistent JSON structures and return appropriate HTTP status codes.

Base URL (local development):

`http://localhost:5000/api/chat`

2. Authentication

👉 **No authentication is required in the current version.**

All endpoints are public for demo/testing.

Future versions may use:

- API keys
- JWT token authentication
- Admin dashboard login

3. Content Type

All requests **MUST** use:

Content-Type: application/json

All responses return JSON.

4. Endpoints Overview

Method	Endpoint	Description
POST	/ask	Main chatbot endpoint
POST	/analyze	Returns intent analysis only
GET	/sports	Returns list of available sports
GET	/health	Backend health check

5. Endpoint Details

5.1 POST /ask

Main Chatbot Endpoint

This is the **core of the entire system**.
It accepts any question from the user and returns an intelligent structured response.

URL

POST /api/chat/ask

Headers

Content-Type: application/json

Request Body

```
{  
  "question": "Where is the basketball class?"  
}
```

Response Structure (Varies by Intent)

A) PE Class Response Example

```
{  
  "response": "Your basketball class is at Basketball Gym. It's on Wednesday from  
08:00 to 10:00 with Coach Wang",  
  "sport": "basketball",  
  "status": "success",  
  "type": "pe",  
  "question_asked": "basketball",  
  "location_details": {  
    "name": "Basketball Gym",  
    "building": null,  
    "floor": null,  
    "room_number": null,  
    "description": "Indoor sports building near East Gate"  
  },  
  "teacher": {  
    "name": "Coach Wang"  
  },  
  "schedule": {  
    "day": "Wednesday",  
    "time": "08:00 - 10:00",  
    "semester": null  
  },  
  "images": [],  
  "maps": {},  
}
```

```
"ai_enhanced": true,  
  
"original_analysis": {  
  
  "intent": "pe_class",  
  
  "sport": "basketball",  
  
  "confidence": 0.92,  
  
  "language": "english"  
  
}  
  
}
```

B) Handbook Response Example

```
{  
  
  "response": "International students must complete 128 credits including mandatory core  
subjects, electives, and PE requirements.",  
  
  "answer": "International students must complete 128 credits including mandatory core  
subjects, electives, and PE requirements.",  
  
  "source": "optimized_handbook",  
  
  "status": "success",  
  
  "type": "handbook",  
  
  "ai_enhanced": false,  
  
  "search_metadata": {  
  
    "confidence": "high",  
  
    "relevance_score": 0.88,  
  
    "total_matches": 3,  
  
    "matched_question": "What are the graduation requirements?",  
  
    "category": "Student Status & Degree Regulations",  
  
    "match_type": "semantic"  
  
  },  
  
},
```

```
"original_analysis": {  
  "intent": "handbook",  
  "handbook_topic": "graduation requirements",  
  "confidence": 0.84,  
  "language": "english"  
}  
}
```

C) Combined Response Example

```
{  
  "response": "Basketball classes at ZJNU usually count as 1 credit. You can check  
your major's graduation plan to see how PE credits fit into your degree progress.",  
  "status": "success",  
  "type": "combined_ai",  
  "ai_generated": true,  
  "original_analysis": {  
    "intent": "combined",  
    "confidence": 0.78  
  }  
}
```

Common Error Responses

Missing JSON Body:

```
{  
  "error": "Content-Type must be application/json",  
  "status": "error"  
}
```

Missing question field:

```
{  
  "error": "Missing 'question' field"  
}
```

Empty message:

```
{  
  "error": "Question cannot be empty"  
}
```

PE Query Error (Database/Logic Failure):

```
{  
  "response": "I'm having trouble finding basketball class information right now. Please  
try again later.",  
  "status": "error",  
  "type": "pe"  
}
```

JSON Handbook Status Error:

```
{  
  "status": "error",  
  "error": "FileNotFoundException: handbook.json not found"  
}
```

5.2 POST /analyze

Intent classification only

This endpoint returns **intent detection**, **confidence**, and **language** WITHOUT generating a full chatbot answer.

Used internally for debugging.

URL

POST /api/chat/analyze

Body

```
{  
  
  "question": "Do I need PE credits to graduate?"  
  
}
```

Response Example

```
{  
  
  "intent": "combined",  
  
  "sport": "PE",  
  
  "handbook_topic": "graduation",  
  
  "confidence": 0.91,  
  
  "language": "english"  
  
}
```

Possible intent values:

- "pe"
- "handbook"
- "combined"
- "unknown"

5.3 GET /sports

Returns list of all available sports classes.

Useful for frontend dropdowns or preloading UI options.

URL

GET /api/chat/sports

Response Example

```
[  
  "basketball",  
  "football",  
  "badminton",  
  "swimming",  
  "volleyball"  
]
```

5.4 GET /health

Health-check endpoint

Used by developers to ensure the backend is running correctly.

URL

GET /api/chat/health

Response Example

```
{  
  "status": "ok",  
  "message": "HandyChat backend is running"  
}
```

6. Response Types Summary

- **PE Response**
 - Structured class info
Location → Teacher → Schedule
- **Handbook Response**
 - JSON-based handbook answer
Optional PDF fallback
- **Combined Response**
 - Merges both above + AI summary
- **Error Response**

Always:

```
{ "error": "description" }
```

7. HTTP Status Codes Used

Code	Meaning	When Used
200	OK	Successful responses
400	Bad Request	Missing fields, invalid JSON
500	Internal Server Error	Any unexpected crash or AI/DB failure

8. Data Sources Used by APIs

8.1 MySQL Database

Used for PE class data:

- Locations
- Teachers
- Sport classes
- Class schedules

8.2 JSON Handbook

Used for fast retrieval of rules & policies.

8.3 PDF Handbook

Used for fallback searches when JSON does not match.

9. Example Full Request (JavaScript)

```
async function ask(question) {  
  
  const response = await fetch('/api/chat/ask', {  
  
    method: 'POST',  
  
    headers: { "Content-Type": "application/json" },  
  
    body: JSON.stringify({ question })  
  
  });  
  
  
  const data = await response.json();  
  
  console.log(data);  
  
}
```

10. Example Full Request (Python)

```
import requests  
  
  
r = requests.post(  
  
  "http://localhost:5000/api/chat/ask",  
  
  json={"question": "Where is badminton class?"}  
  
)  
  
  
print(r.json())
```

11. Future API Extensions

- /api/admin/add_sport
- /api/admin/update_schedule
- /api/chat/voice (future)
- JWT-protected routes

- WebSocket for real-time chatbot typing indicator

12. Conclusion

This API documentation provides a complete guide for:

- Frontend developers
- Integration with mobile apps
- Debugging and testing
- Future improvements

The HandyChat API is lightweight, clear, and easy to extend — ideal for university systems and student support tools.